UNIVERSITY OF BRISTOL

School of Mathematics

FINANCIAL MATHEMATICS

MATH35400

(Paper code MATH–35400)

May/June 2023   2 hours 30 minutes

This paper contains 4 questions. All answers will be used for assessment.

- Calculators of an approved type (permissible for A-Level examinations) are permitted.
- Candidates may bring four sheets of A4 notes written double-sided into the examination. Candidates must insert these into their answer booklet(s) for collection at the end of the examination.
- On this examination, the marking scheme is indicative and is intended only as a guide to the relative weighting of the questions.

You may find the following results useful:

- The *put-call-parity* states that

$$S_t + P_t - C_t = Ke^{-r(T-t)}$$

  where S_t is the price of the asset at time point t, $Ke^{-r(T-t)}$ the discounted value of the bond and C_t and P_t the prices of European call and put options at time t.

- *Itô's lemma* for the derivative of a function of an Itô process with stochastic differential $dX_t = b_t dt + \sigma_t dW_t$:

$$df = \left(f_t + b_t f_x + \frac{1}{2}\sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t.$$

Do not turn over until instructed.

1. **(25 marks)** Consider a single-period model with N risky securities and a finite probability space $\Omega = \{\omega_1, \omega_2, \omega_3\}$. For each state $\omega \in \Omega$ and time point $t = 0, 1$ let $S_n(t)(\omega)$ denote the price of the n-th security at time point t in state ω. We assume that there is a bank account with interest rate $r = 0.1$.

   (a) **(6 marks)** Let us first assume that $N = 1$ and the price process is given by the following table:

$S_1(0)$	$S_1(1)(\omega_1)$	$S_1(1)(\omega_2)$	$S_1(1)(\omega_3)$
11	11	16.5	8.8

   Specify the value and gains processes V_t and G for an arbitrary trading strategy $H = (H_0, H_1)$ in this single-period model as well as their discounted versions V_t^* and G^*. As usual H_0 corresponds to the bank account and H_1 to the risky security.

   (b) **(8 marks)** Determine all risk-neutral probability measures for the model given in (a).

   (c) **(5 marks)** Assume now that $N = 2$ and the price process is given by the following table:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	11	11	16.5	8.8
2	7	12.1	6.6	7.7

   Determine all risk-neutral probability measures. Then use the risk-neutral valuation principle to find the time 0 prices for European call options with strike price $K = 11$ on (i) the first security and (ii) the second security.

   (d) **(6 marks)** Assume now that $N = 3$ and the price processes are given by the following table:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	11	11	16.5	8.8
2	7	12.1	6.6	7.7
3	5	6.6	3.3	8.8

   By comparison with the model in (c) for $N = 2$ explain why there must be arbitrage opportunities.

   Then specify a trading strategy $H = (H_1, H_2, H_3)$ which is an arbitrage opportunity and calculate its discounted gain for every state ω.

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2. (**25 marks**) Consider a financial model with $T = 2$ periods where the stock prices are given in the table below and the interest rate is $r = 0.2$.

State	$t = 0$	$t = 1$	$t = 2$
ω_1	10	13.2	17.42
ω_2	10	13.2	14.26
ω_3	10	10.8	14.26
ω_4	10	10.8	11.67

(a) **(7 marks)** Determine all risk-neutral probability measures $\mathbb{Q} = (q_1, q_2, q_3, q_4)$ where $q_i = \mathbb{Q}(\omega_i)$. Which special case of multi-period model is this model? Explain briefly why.

(b) **(8 marks)** Calculate the price of a European call option with maturity date $T = 2$ and strike price $K = 12$ at times $t = 0, 1, 2$. What is the price of an American call option with the same maturity date and the same strike price at times $t = 0, 1, 2$?

(c) **(10 marks)** Let X^a be the American call option with maturity date $T = 2$ and payoff process $g(S_t, t) = (S_t - K_t)^+$, where the variable strike price K_t satisfies $K_0 = 9, K_1 = 9.9$ and $K_2 = 12$. Determine the price of this option at times $t = 0, 1, 2$ and the optimal exercise strategy $\tau(0)$. Then compute a replicating strategy.

Continued...

3. (**25 marks**)

(a) Consider the Cox-Ross-Rubinstein model which consists of a bank account with constant interest rate $r > 0$ and one risky asset with price process S_t defined by

$$S_t = S_0 u^{N_t} d^{t-N_t}, \quad t = 1, 2, \dots, T.$$

Here u and d are real numbers such that $0 < d < u$, $S_0 > 0$, and $N_t = X_1 + \cdots + X_t$ is the random walk process associated with a Bernoulli process X_t with parameter p, where $0 < p < 1$.

i. **(4 marks)** State the distribution of $N_{t+u} - N_t$ for $u, t > 0$ as well as its mean and variance. Then show that $N_t - tp$ is a martingale.

ii. **(3 marks)** Show that the process Z_t defined by

$$Z_t = \exp(\sigma N_t + at) \tag{1}$$

is a martingale if $e^a(1 - p + pe^\sigma) = 1$.
[Hint: The moment-generating function of a binomial random variable W with parameters m and p is $M_W(t) = \mathbb{E}(\exp(tW)) = (1 - p + pe^t)^m$.]

iii. **(4 marks)** Show that the discounted price process S_t^* can be expressed in the form (1), i.e. can be written as

$$S_t^* = \exp(\sigma N_t + at)$$

for suitable σ and a. Then show that S_t^* is a martingale, if $p = (1+r-d)/(u-d)$.

iv. **(2 marks)** State the definition of a martingale measure in the multi-period model. What condition does a multi-period model need to satisfy, so that there are no arbitrage opportunities?

v. **(2 marks)** Explain why there are no arbitrage opportunities in the model, if $p = (1 + r - d)/(u - d)$.

(b) An investor proposes to replace the famous Black-Scholes model by the following model for the stock:

$$dS_t = rS_t dt + \sigma dW_t,$$
$$S_0 = x$$

where $(W_t)_{t \geq 0}$ is standard Brownian motion, r is the constant interest rate and x and σ are constants.

i. **(3 marks)** Use Itô's lemma to show that

$$d(e^{-rt} S_t) = \sigma e^{-rt} dW_t.$$

for S_t satisfying the model above.

ii. **(3 marks)** Show that

$$S_t = xe^{rt} + \int_0^t e^{r(t-s)} \sigma dW_s.$$

iii. **(3 marks)** Determine mean and variance of S_t.

iv. **(1 marks)** Assume that S_t has a normal distribution – why should the proposal be rejected?

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4. (**25 marks**) Let $(W_t)_{t\geq 0}$ be the standard Brownian motion. The Black-Scholes model is a financial market model in continuous time with consists of a riskless bond with constant interest rate r and a stock whose price S_t is modelled as

$$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t).$$

The Black-Scholes formula for the price at time t of a European call option with exercise price K and maturing at time T is given by

$$C(t) = S_t \Phi(d_1(S_t, T-t)) - Ke^{-r(T-t)}\Phi(d_2(S_t, T-t))$$

where the functions $d_1(s,t)$ and $d_2(s,t)$ are given by

$$d_1(s,t) = \frac{\log(s/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}} \quad \text{and} \quad d_2(s,t) = \frac{\log(s/K) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

and Φ is the standard normal cumulative distribution function.

(a) **(3 marks)** Find the time 0 price of a European call option on a stock when the stock price is 50, the strike price is 60, the risk-free interest rate is 3% per annum, the volatility is 0.3 per annum and the time to maturity is six months. *[Provide a numerical answer, you might want to use the table at the end of this exam paper.]*

(b) **(5 marks)** What is the probability that a European call option with strike price K and maturity T expires out of the money, i.e. has zero value? Write the probability as $\Phi(D)$ where D is an expression that you need to determine. Evaluate this probability for the option in (a) assuming that $\mu = 0.05$.

(c) **(4 marks)** In the above stock price model for S_t, what are the usual names for the parameters μ and σ? Show that

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

(d) **(2 marks)** Let $\mathbb{Q}$ be the equivalent martingale measure. Define the process $\overline{W}_t$ such that it is a standard Brownian motion under $\mathbb{Q}$. Write S_t in terms of $\overline{W}_t$.

(e) **(6 marks)** Use the risk-neutral valuation principle to calculate the time t price $g(t, S_t)$ of a European option with payoff $(S_T - K)^2$ at time T for some $K > 0$.

(f) **(5 marks)** Show that the function $g(t,x)$ in the previous question satisfies the Black-Scholes-Merton differential equation

$$g_t + rxg_x + \frac{\sigma^2}{2}x^2 g_{xx} = rg.$$

End of examination.

Probabilities of a standard normal distribution

The next tables shows probabilities from a standard normal distribution. Each cell contains the value of $\Phi(x_1 + x_2)$ where x_1 and x_2 are the labels of the corresponding row and column.

Φ	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990